RAJALAKSHMI ENGINEERING COLLEGE

Kanmani S 251801129-2025-AIDS

K2

Started on	Thursday, 24 September 2026, 8:24 AM
State	Finished
Completed on	Thursday, 24 September 2026, 8:26 AM
Time taken	1 min 33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int binarySearch(int a[], int low, int high, int x)
{
    if (low > high)
        return -1;

    int mid = (low + high) / 2;

    if (a[mid] == x)
        return mid;

    if (a[mid] > x)
        return binarySearch(a, low, mid - 1, x);

    return binarySearch(a, mid + 1, high, x);
}

int findPair(int a[], int low, int high, int x, int *e1, int *e2)
{
    if (low >= high)
        return 0;
    int mid = (low + high) / 2;
    if (findPair(a, low, mid, x, e1, e2))
        return 1;
    if (findPair(a, mid + 1, high, x, e1, e2))
        return 1;
    for (int i = low; i <= mid; i++)
    {
        int complement = x - a[i];

        int pos = binarySearch(a, mid + 1, high, complement);

        if (pos != -1)
        {
            *e1 = a[i];
            *e2 = a[pos];
            return 1;
        }
    }

    return 0;
}

int main()
{
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
    {
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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